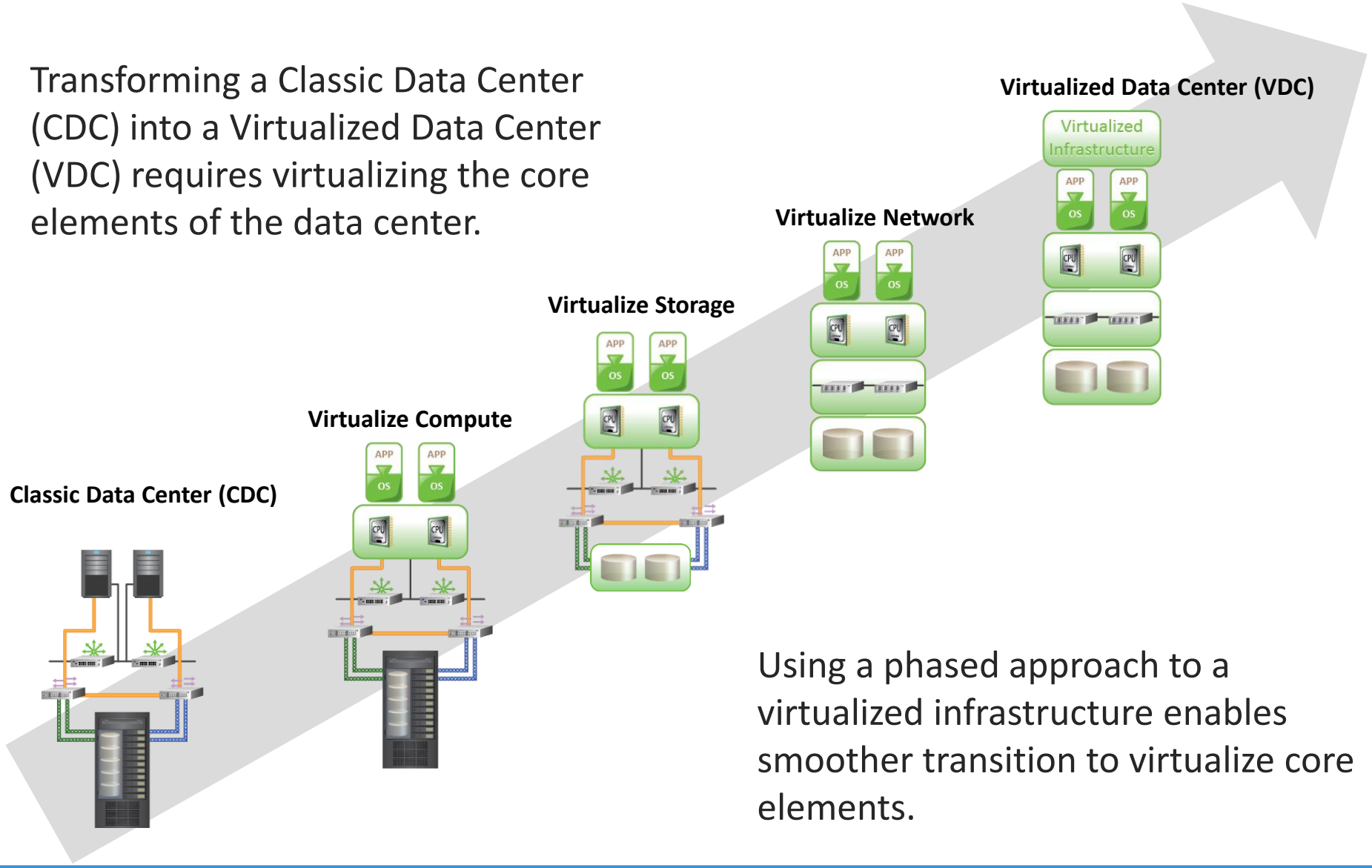
THE VIRTUALISED DATA CENTRE

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Virtualized Data Center

Transforming a Classic Data Center (CDC) into a Virtualized Data Center (VDC) requires virtualizing the core elements of the data center.



Using a phased approach to a virtualized infrastructure enables smoother transition to virtualize core elements.

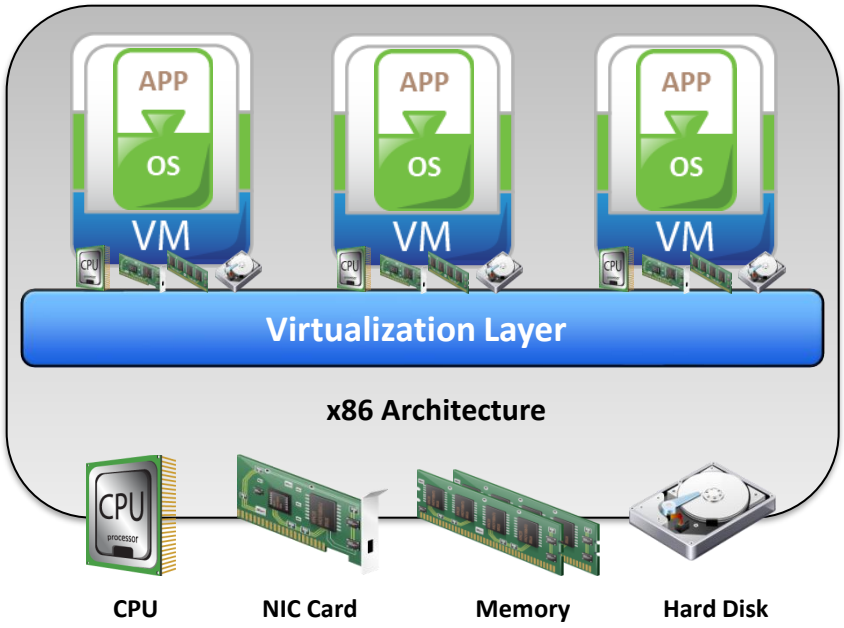
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Compute Virtualization

Compute Virtualization

It is a technique of masking or abstracting the physical compute hardware and enabling multiple operating systems (OSs) to run concurrently on a single or clustered physical machine(s).

- Enables creation of multiple virtual machines (VMs), each running an OS and application
  - VM is a logical entity that looks and behaves like physical machine
- Virtualization layer resides between hardware and VMs
  - Also known as hypervisor
- VMs are provided with standardized hardware resources



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Benefits of Compute Virtualization

- Server consolidation
- Isolation
- Encapsulation
- Hardware independence
- Reduced cost

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Resource Management

Resource management

A process of allocating resources from physical machine or clustered physical machines to virtual machines (VMs) to optimize the utilization of resources.

- Goals of resource management
  - Controls utilization of resources
  - Prevents VMs from monopolizing resources
  - Allocates resources based on relative priority of VMs
- Resources must be pooled to manage them centrally

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Resource Pool

Resource pool

It is a logical abstraction of aggregated physical resources that are managed centrally.

- Created from a physical machine or cluster
- Administrators may create child resource pool or virtual machine (VM) from the parent resource pool
- Reservation, limit, and share are used to control the resources consumed by resource pools or VMs

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Resource Pool Example



Standalone Physical Machine – Machine 1
Parent Pool
CPU = 3.5 GHz
Memory = 64GB

Engineering Pool (Child Pool)
CPU = 1 GHz
Memory = 16GB

Finance Pool (Child Pool)
CPU = 1 GHz
Memory = 16GB

Marketing-Production VM
CPU = 500 MHz
Memory = 8GB

Engineering-Test VM
CPU = 500 MHz
Memory = 8GB

Engineering-Prod VM
CPU = 500 MHz
Memory = 8GB

Finance-Test VM
CPU = 500 MHz
Memory = 8GB

Finance-Production VM
CPU = 500 MHz
Memory = 8GB

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Share, Limit, and Reservation

- Parameters that control the resources consumed by a child resource pool or a virtual machine (VM) are as follows:
  - Reservation
    - Amount of CPU and memory reserved for a VM or a child resource pool
  - Limit
    - Maximum amount of CPU and memory a VM or a child resource pool can consume
  - Share
    - Amount of CPU or memory resources a VM or a child resource pool can have with respect to its parent's total resources

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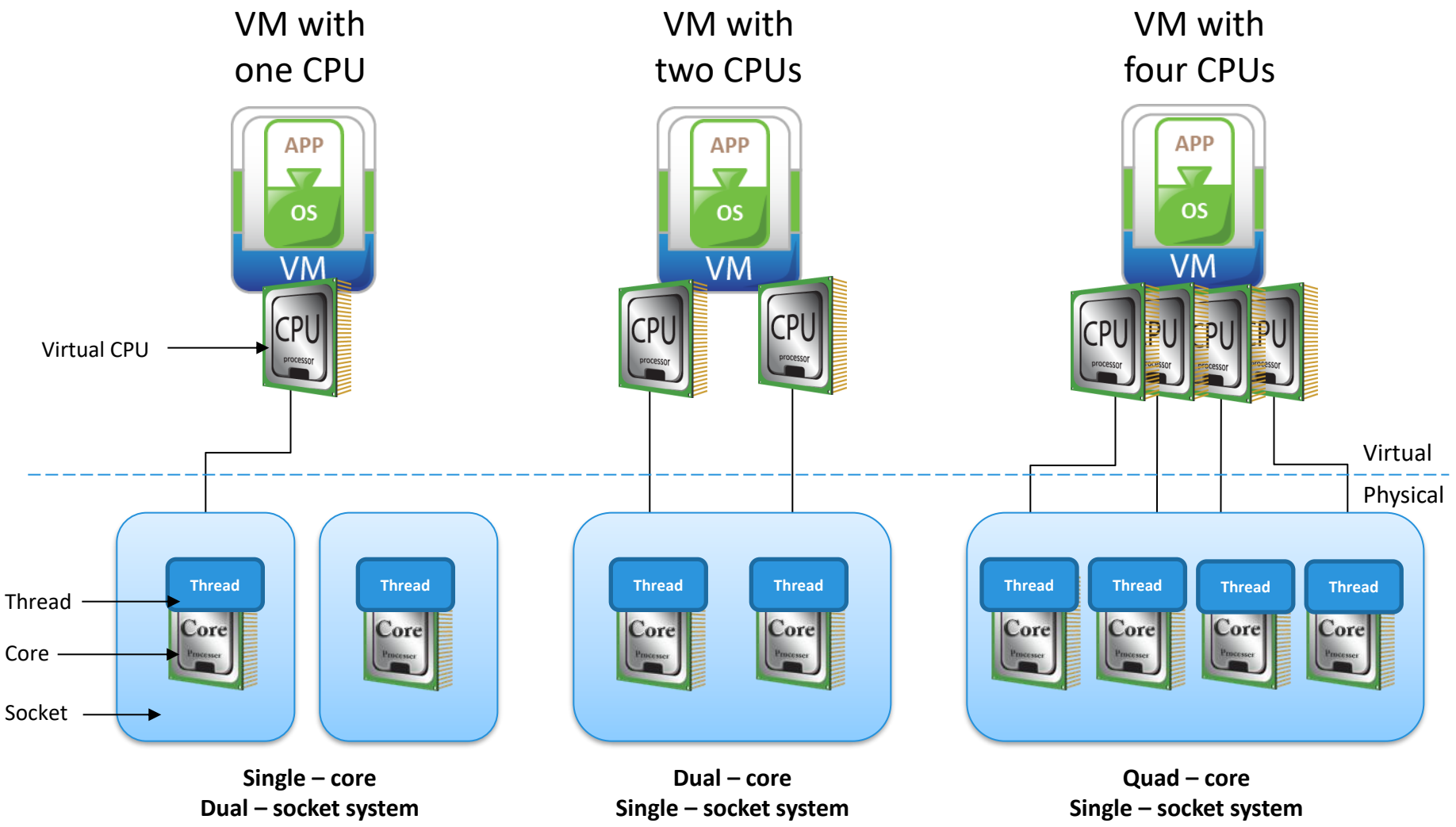
Optimizing CPU Resources

- Modern CPUs are equipped with multiple cores and hyper-threading
  - Multi-core processors have multiple processing units (cores) in a single CPU
  - Hyper-threading makes a physical CPU appear as two or more logical CPUs
- Allocating a CPU resource efficiently and fairly is critical
- Hypervisor schedules virtual CPUs on the physical CPUs
- Hypervisors support multi-core, hyper-threading, and CPU load-balancing features to optimize CPU resources

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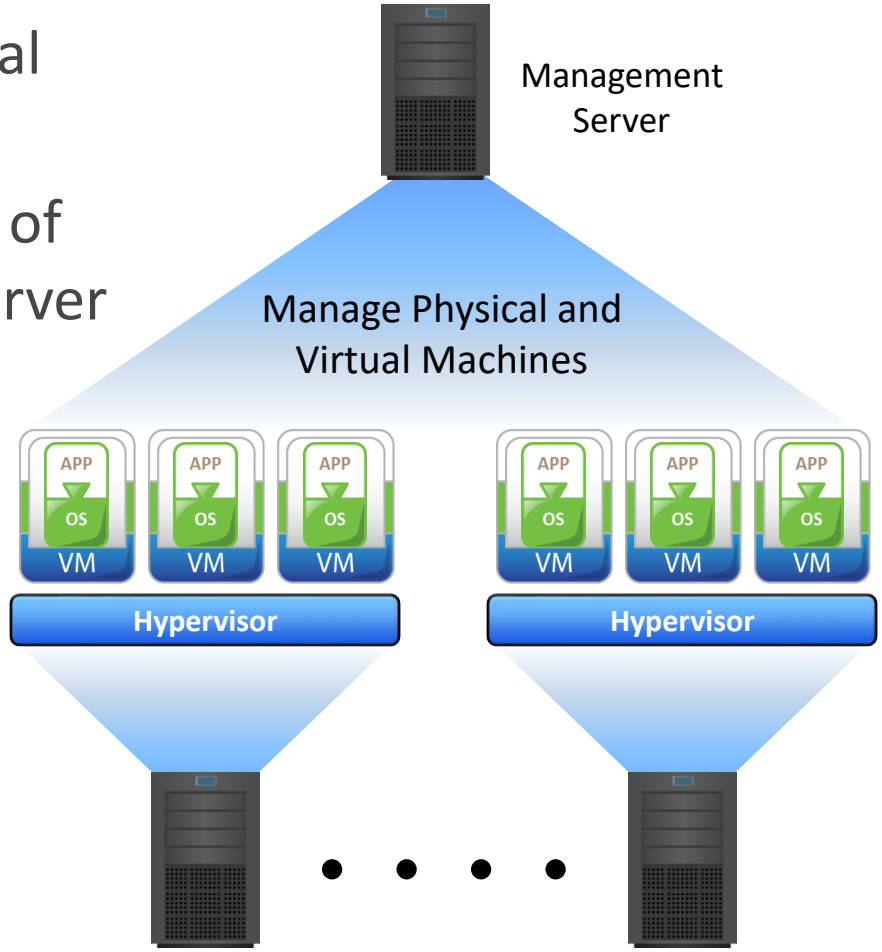
Multi-core Processors



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Resource Management Tool

- Provides ability to manage physical machines running hypervisor
- Enables centralized management of resources from a management server
- Enables pooling of resources and allocates capacity to VMs
  - Communicates with hypervisors to perform management
- Provides operational automation



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Storage Virtualization

Storage virtualization

It is the process of masking the underlying complexity of physical storage resources and presenting the logical view of these resources to compute systems.

- Logical to physical storage mapping is performed by virtualization layer
- Virtualization layer abstracts the identity of physical storage devices
  - Creates a storage pool from multiple, heterogeneous storage arrays
- Virtual volumes are created from the storage pools and are assigned to the compute system

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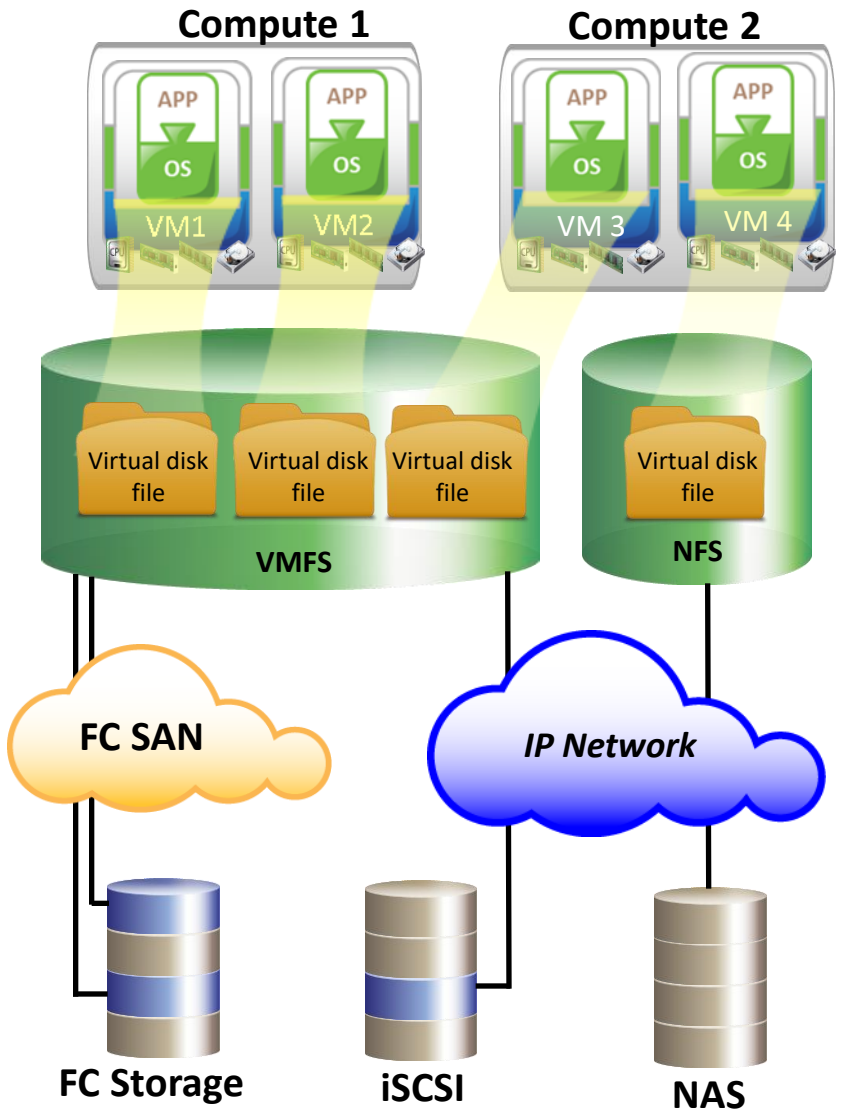
Benefits of Storage Virtualization

- Adds or removes storage without any downtime
- Increases storage utilization thereby reducing TCO
- Provides non-disruptive data migration between storage devices
- Supports heterogeneous, multi-vendor storage platforms
- Simplifies storage management

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Storage for Virtual Machines

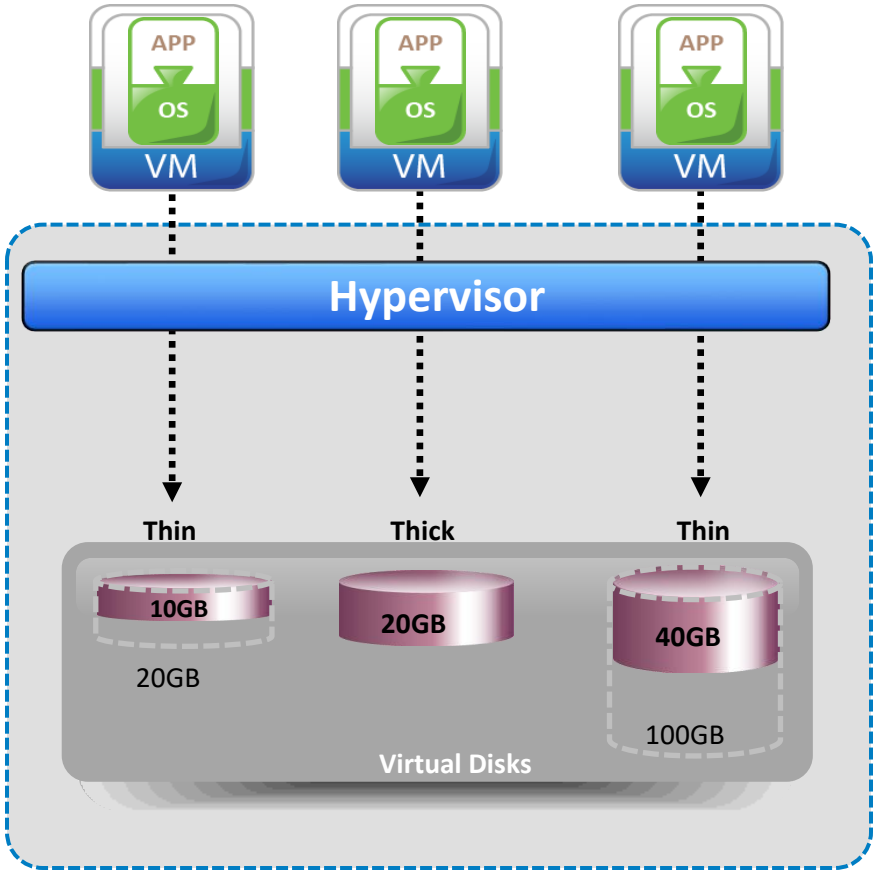
- VMs are stored as set of files on storage space available to hypervisor
- 'Virtual disk file' represents a virtual disk used by a VM to store its data
- Size of virtual disk file represents storage space allocated to virtual disk
- VMs remain unaware of
  - Total space available to the hypervisor
  - Underlying storage technologies



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Virtual Provisioning at Compute

- Hypervisor performs virtual provisioning to create virtual disks for VMs
  - Virtual machine sees full logical disk size at all times
- Hypervisor allocates storage space to the virtual disk only when VM requires storage space
  - Eliminates the need to overprovision virtual disks



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Virtual Provisioning Benefits

- Reduces administrative overhead
- Improves capacity utilization
- Reduces cost
- Reduces downtime

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Network Virtualization

Network Virtualization

It is a process of logically segmenting or grouping physical network(s) and making them operate as single or multiple independent network(s) called "Virtual Network(s)".

- Enables virtual networks to share network resources
- Allows communication between nodes in a virtual network without routing of frames
- Enforces routing for communication between virtual networks
- Restricts management traffic, including 'Network Broadcast', from propagating to other virtual network
- Enables functional grouping of nodes in a virtual network

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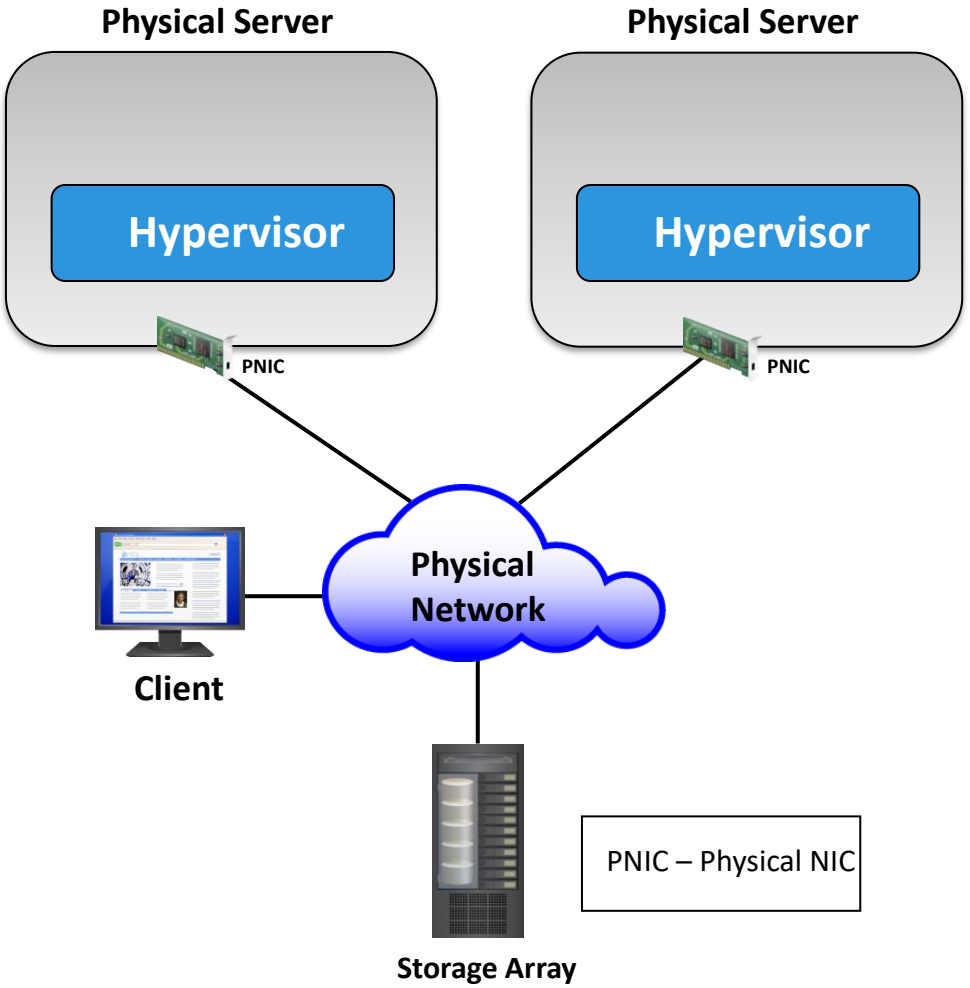
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Network Virtualization in VDC

- Involves virtualizing physical and VM networks

Physical Network

- Consists of following physical components:
  - Network adapters, switches, routers, bridges, repeaters, and hubs
- Provides connectivity
  - Among physical servers running hypervisor
  - Between physical servers and clients
  - Between physical servers and storage systems

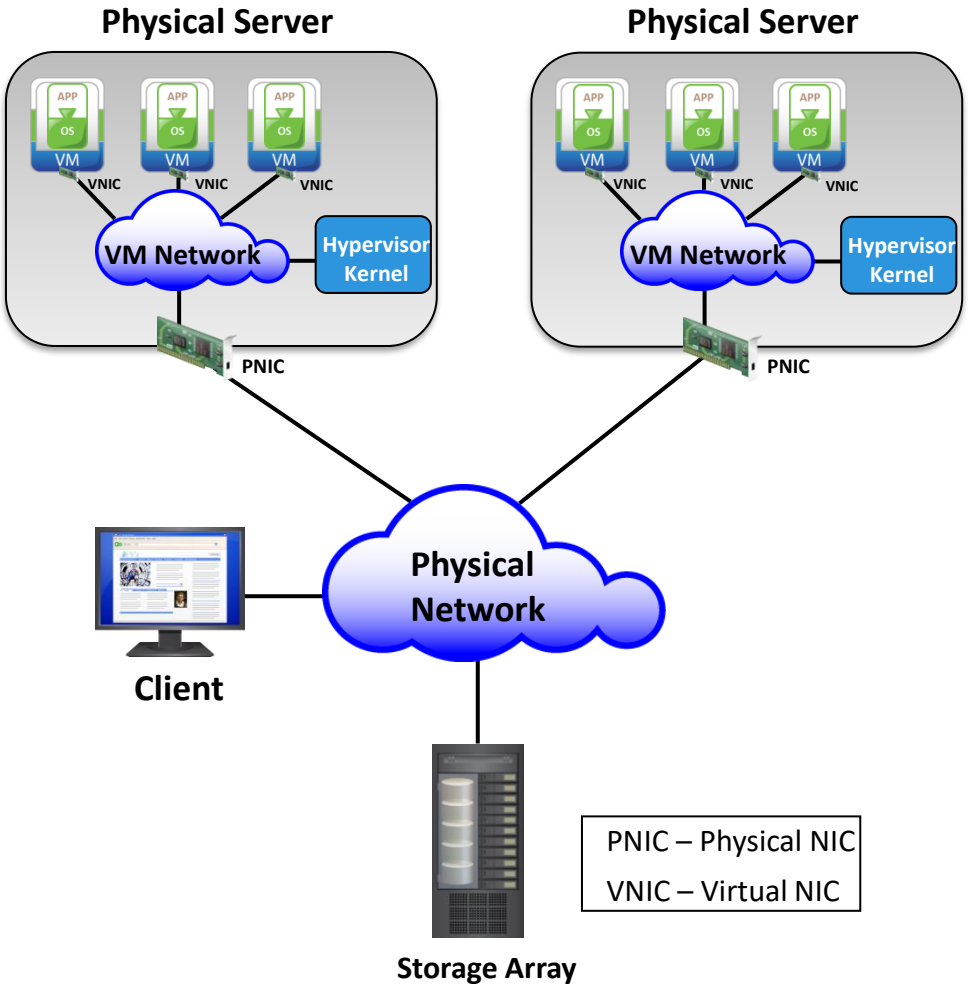


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Network Virtualization in VDC (contd.)

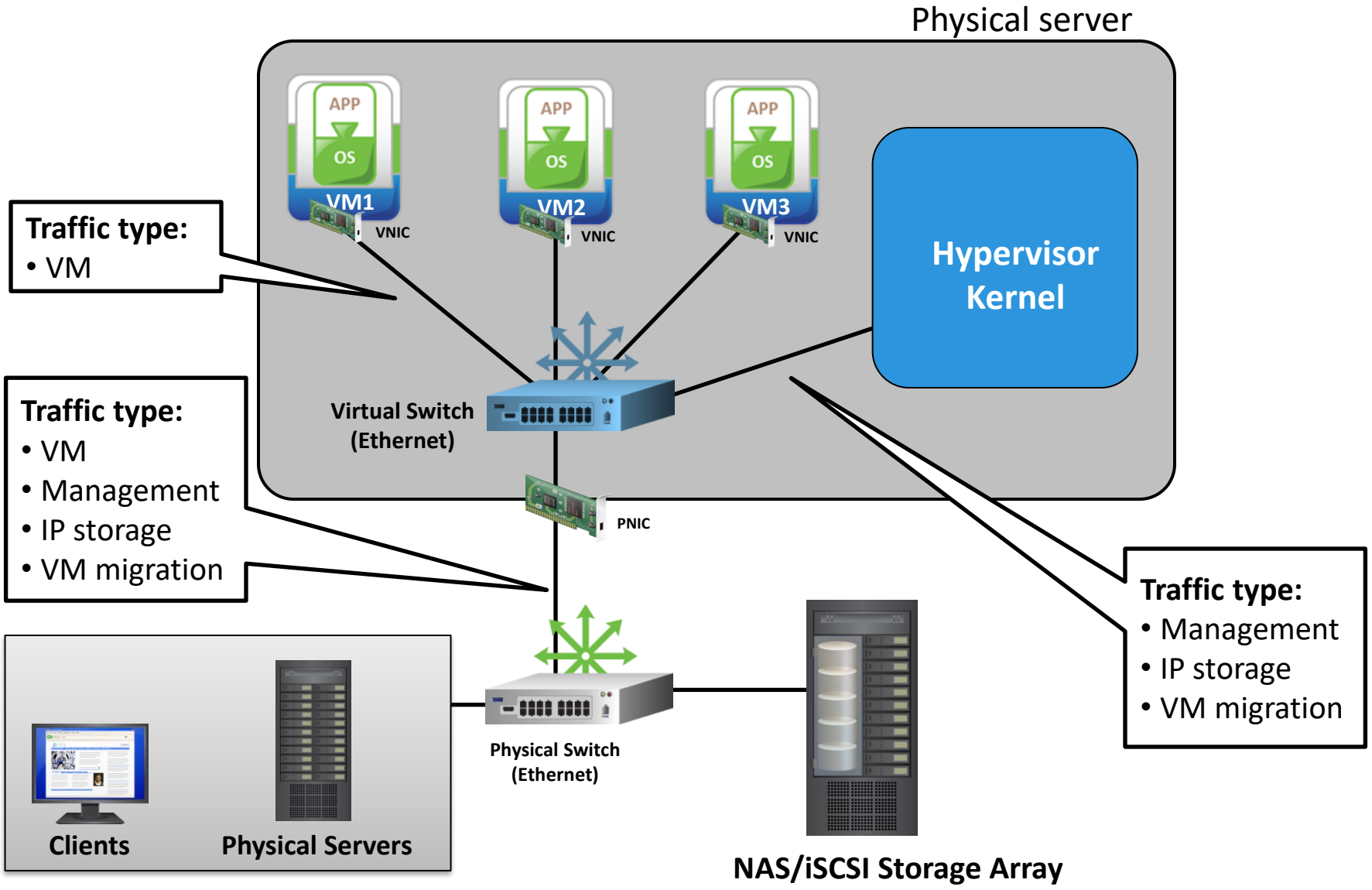
VM Network

- Resides inside physical server
- Consists of logical switches called “virtual switches”
- Provides connectivity among VMs inside a physical server
- Provides connectivity to Hypervisor kernel
- Connects to physical network



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Network Connectivity and Traffic Flow: Example



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Benefits of Network Virtualization

Benefit	Description
Enhances security	• Restricts access to nodes in a virtual network from another virtual network • Isolates sensitive data from one virtual network to another
Enhances performance	• Restricts network broadcast and improves virtual network performance
Improves manageability	• Allows configuring virtual networks from a centralized management workstation using management software • Eases grouping and regrouping of nodes
Improves utilization and reduces CAPEX	• Enables multiple virtual networks to share the same physical network, which improves utilization of network resource • Reduces the requirement to setup separate physical networks for different node groups

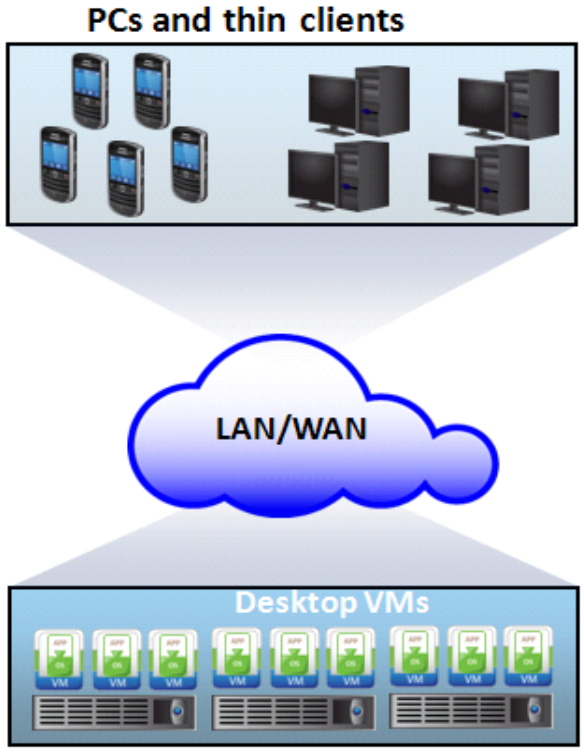
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Desktop Virtualization

Desktop Virtualization

Technology which enables detachment of the user state, the Operating System (OS), and the applications from endpoint devices.

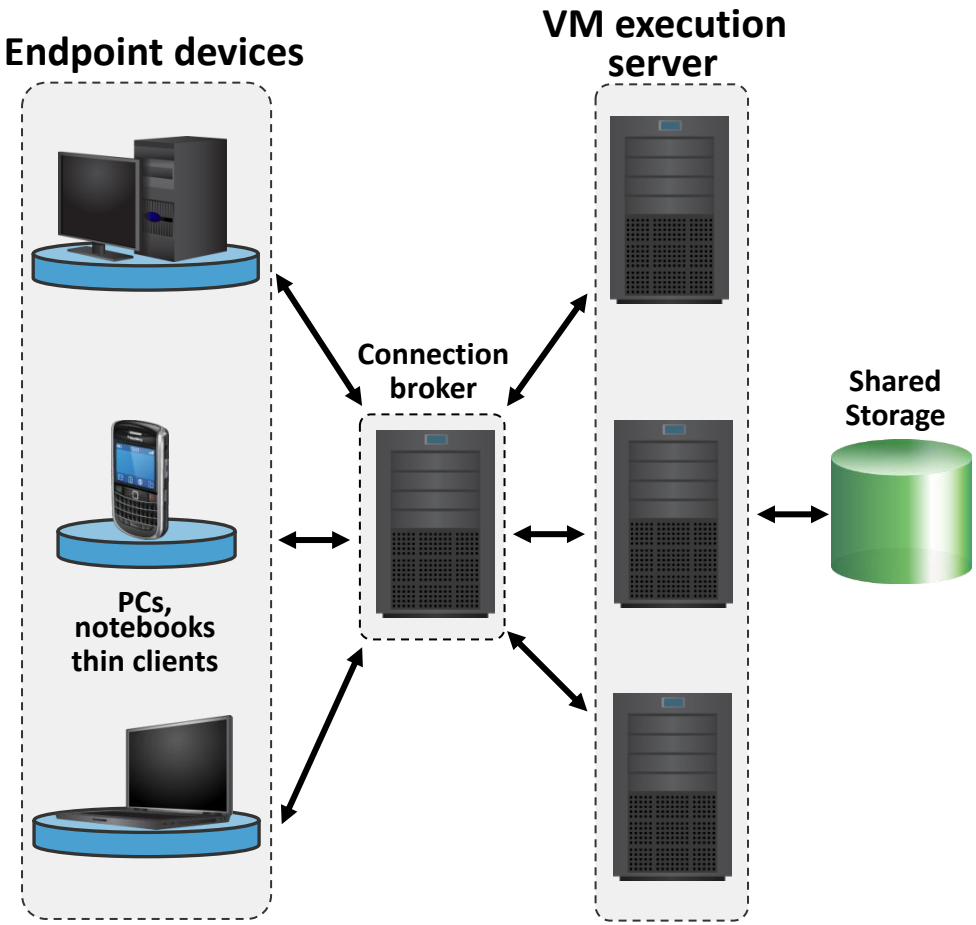
- Enables organizations to host and centrally manage desktops
  - Desktops run as virtual machines within the VDC
    - They may be accessed over LAN/WAN
  - Endpoint devices may be thin clients or PCs



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Virtual Desktop Infrastructure (VDI): Components

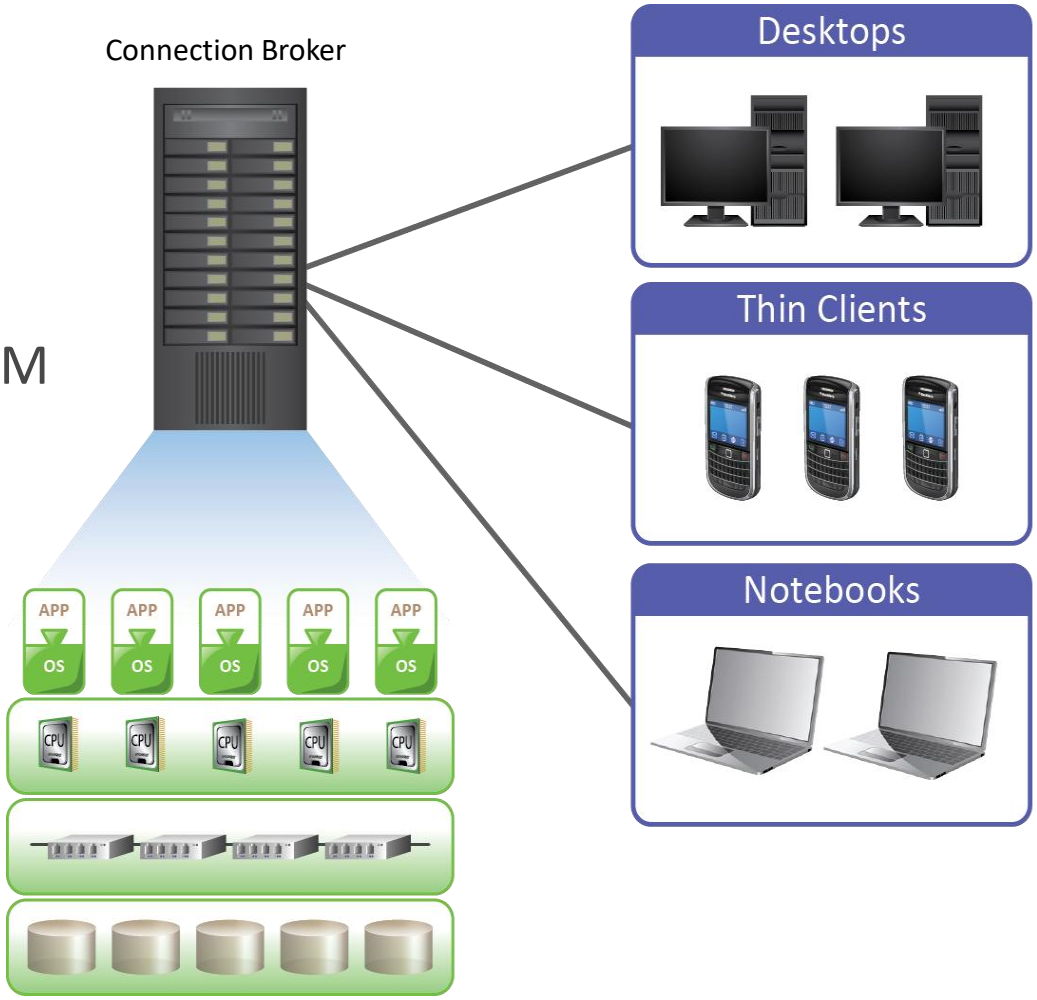
- Endpoint devices
- VM hosting/execution servers
- Connection Broker



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Connection Broker

- It is responsible for establishing and managing the connection
  - Between the endpoint device and the desktop VM



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Application Virtualization

Application Virtualization

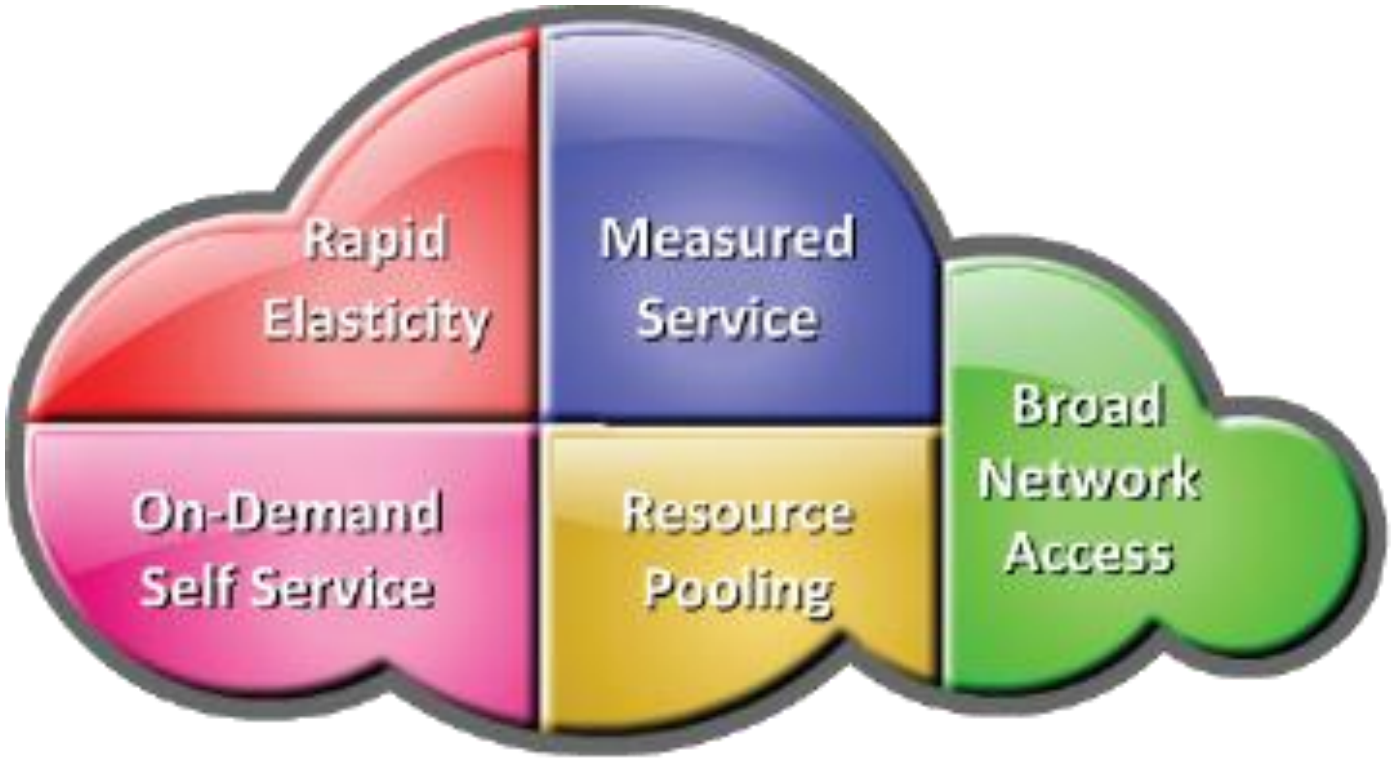
It is the technique of presenting an application to an end user without any installation, integration, or dependencies on the underlying computing platform

- Allows application to be delivered in an isolated environment
  - Aggregates Operating System (OS) resources and the application into a virtualized container
  - Ensures integrity of Operating System (OS) and applications
  - Avoids conflicts between different applications or different versions of the same application

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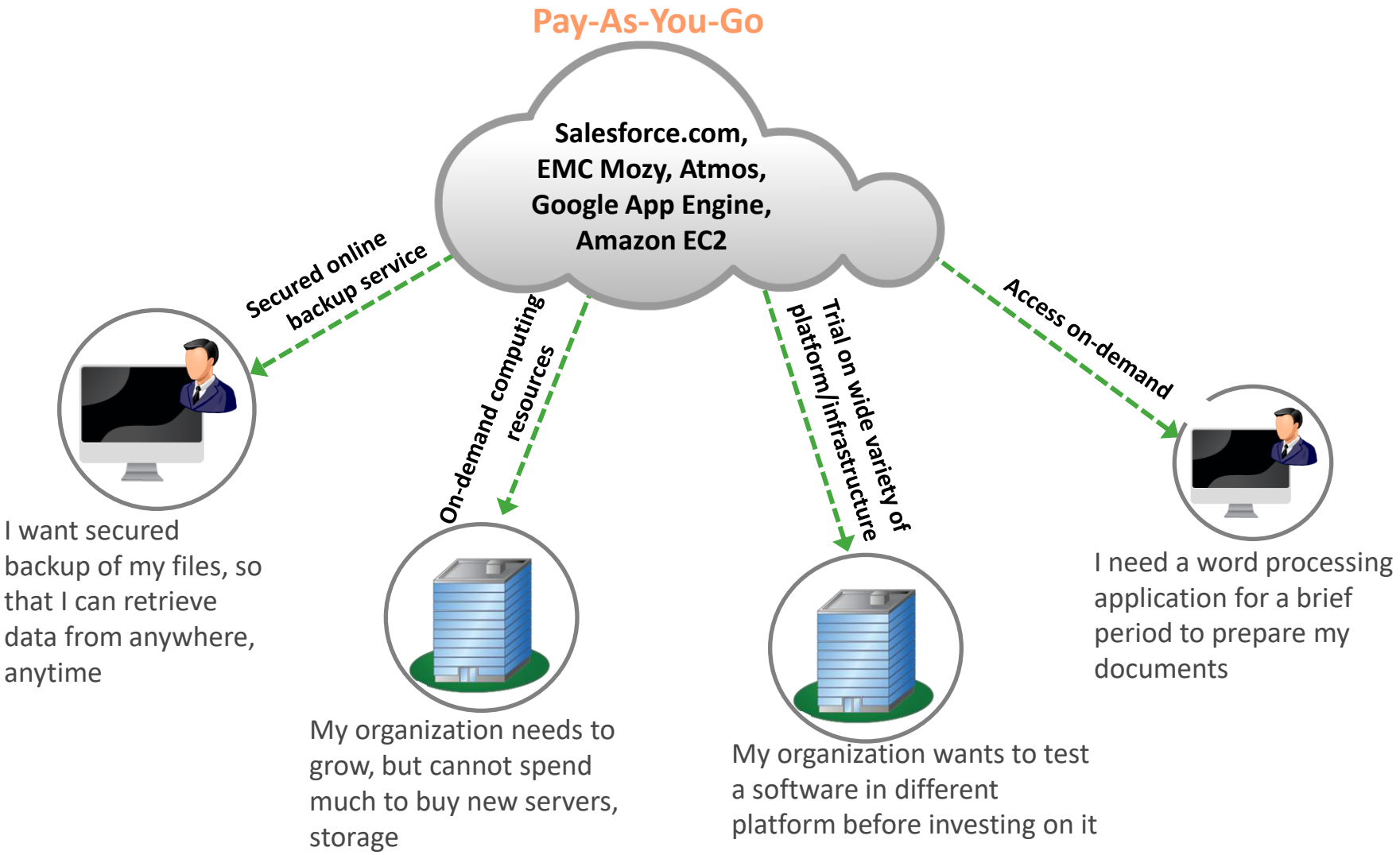
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Cloud Computing: Essential Characteristics



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Cloud Offering Examples



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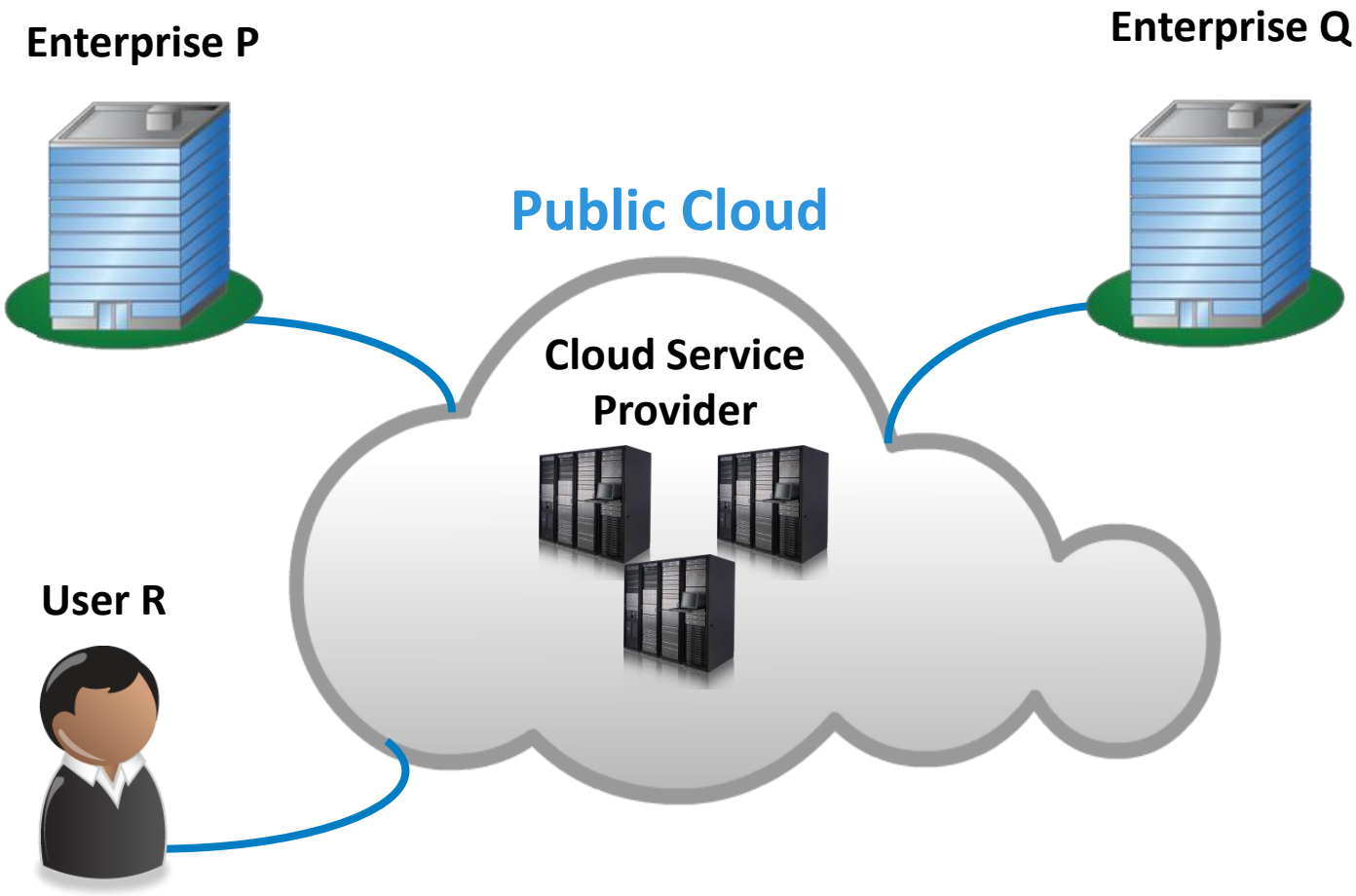
Cloud Computing Benefits

Benefit	Description
Reduced IT Cost	• Avoids the up-front capital expenditure
Business agility support	• Provides the ability to add new resources quickly
Flexible scaling	• Scales up and down easily and instantly, based on demand
High availability	• Ensures application availability at varying levels, depending on policy and priority of the application
Less energy consumption	• Enables organizations to reduce power consumption and space usage

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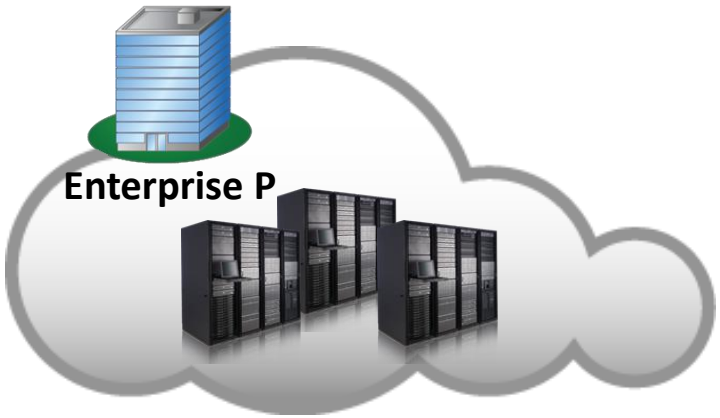
Cloud Deployment Model – Public Cloud



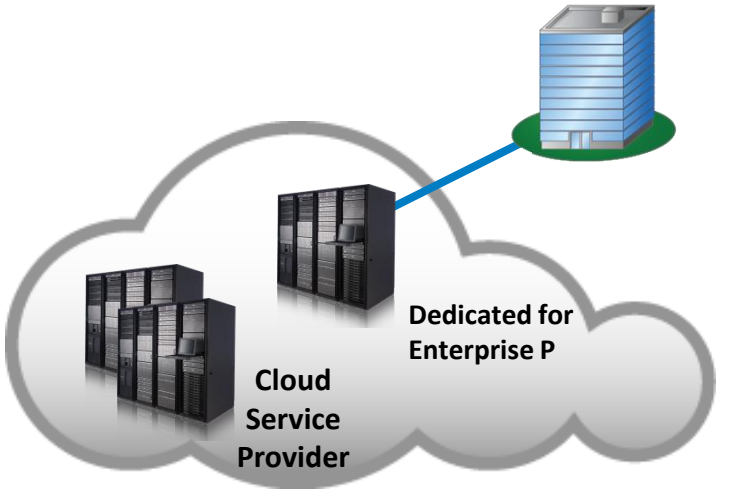
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Cloud Deployment Model – Private Cloud

On-premise Private Cloud

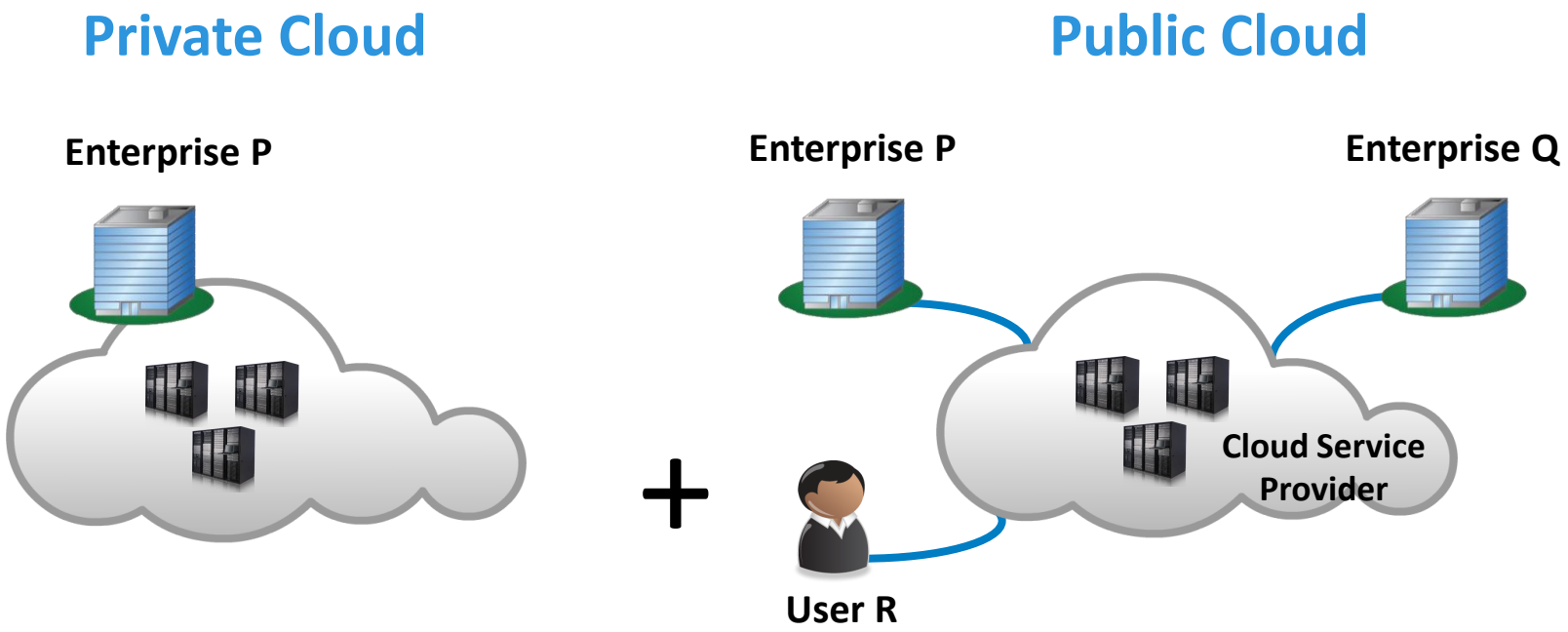


Externally hosted Private Cloud



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Cloud Deployment Model – Hybrid Cloud



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Economics of Cloud

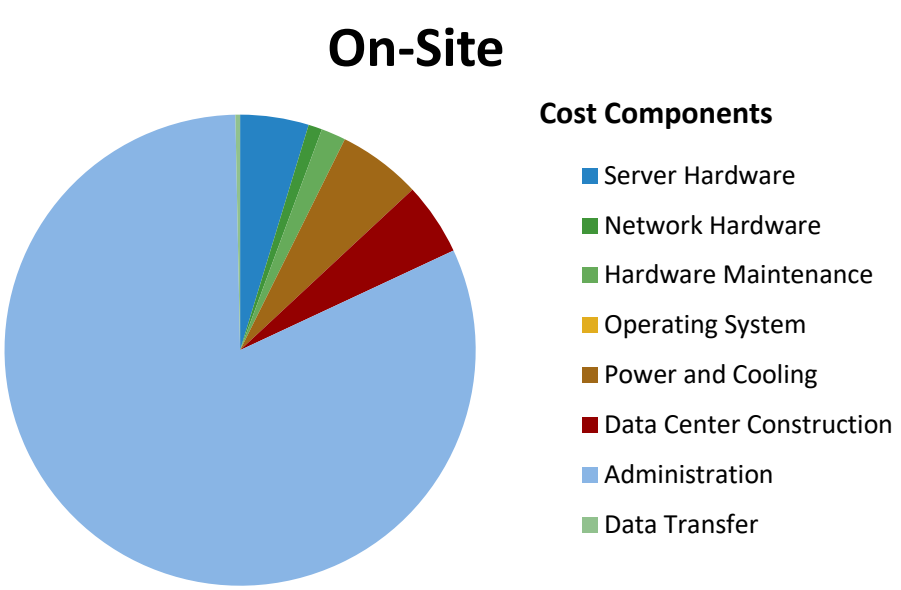
- Cloud has changed the economics of IT
- Cloud enables to move from a CAPEX to an OPEX model
- Cloud provides the following key cost savings
  - Infrastructure cost
  - Management cost
  - Power and energy cost

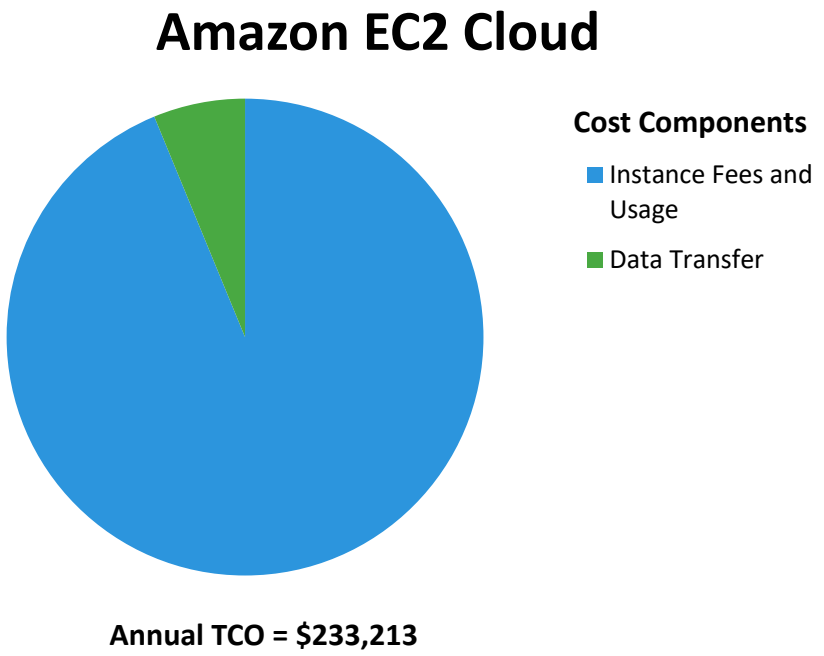
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Economics of Cloud Example: On-Site Vs. Cloud

Buying 1000 Servers (On-Site) Vs. Hiring 1000 server instances (Cloud)



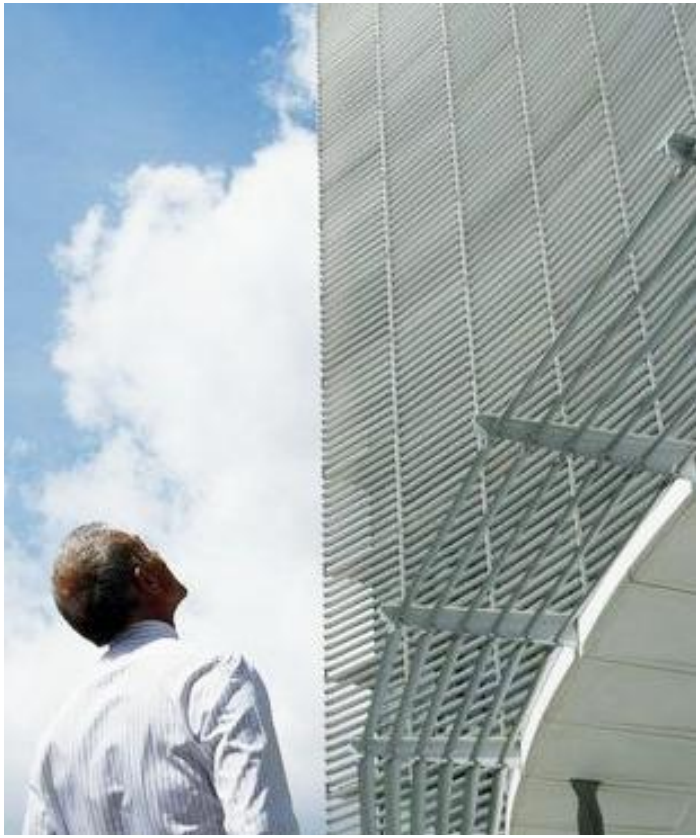


Source: Amazon Web Services: The Economics of the AWS Cloud vs. Owned IT Infrastructure, Dec 2009

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Cloud Challenges – Consumer's Perspective

- Security and Regulation
  - Consumers are indecisive to transfer control of sensitive data
  - Regulation may prevent organizations to use Cloud services
- Network latency
  - Real time applications may suffer due to network latency and limited bandwidth
- Supportability
  - Legacy or Custom applications may not be compatible with Cloud platform
- Interoperability
  - Lack of standardization across Cloud-based platforms



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Cloud Challenges – Provider's Perspective

- Service warranty and service cost
  - Resources must be kept ready to meet unpredictable demand
  - Hefty penalty, if SLAs are not fulfilled
- Huge numbers of software to manage
  - Huge number of applications and platform software to purchase
  - ROI is unpredictable
- No standard Cloud access interface
  - Cloud customers want open APIs
  - Need agreement among Cloud providers for standardization

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